Topological analyses of Exp 4 / 4b: goal and procedure

GPT-2 small, 200 random rows $\times$ 11 country–capital pairs. $D(d)$ = change in the final token's attention to position τ when a country sits there instead of a control token, at distance $d = 32 - \tau$ ($d = 1..31$). Final token = its capital (own), a wrong capital, or a random token. Each analysis was pre-registered in `exp4/PREDICTIONS_tda`k`.md` before it ran.

1 · Persistent homology of head profiles

`exp4/tda1_heads.py`

Goal. Test whether the six head types set by hand thresholds in Exp 4 are real, persistent clusters of heads, and whether continuous families (loops) exist among heads.

Procedure

1. Make one point per head (144): concatenate its $D(d)$ curves for the country arm under a random, own and wrong final token, plus the placebo-word arm, in both filler types (248 numbers).
2. Build a distance matrix between heads: Euclidean on raw values (main); correlation and z-scored distances as robustness checks.
3. Run Vietoris–Rips persistent homology (ripser): H0 bars are clusters, H1 bars are loops.
4. Resample the 200 rows 500 times. Call a feature real only if it beats the bootstrap noise band (Fasy et al.), and score each cluster by how often it recurs (Jaccard ≥ 0.75 = stable). Compare against two null models.
5. Compare the clusters to the hand-set labels (adjusted Rand index) and check whether the four same-country heads group together.

2 · Mapper over depth

`exp4/tda2_mapper.py`

Goal. Test whether a head's range (local vs global response to a country) changes smoothly with layer or splits into branches, and test the post hoc "layers 2–3 local, 5–8, 9–11 global" grouping as a hypothesis.

Procedure

1. One point per head (144): its 6 country-arm curves (own, wrong, random $\times$ common, rare), 186 numbers, rescaled to shape $(v/(\|v\| + \lambda))$ so magnitude does not dominate.
2. Use the layer index as the lens and cover layers 0–11 with overlapping windows (main: 4 layers wide, step 2).
3. Within each window, cluster heads by single linkage at a distance cutoff ε set from the data.
4. Draw the graph: one node per cluster per window; join two nodes when they share a head. Read off branches and components, and test the six pre-registered hypotheses (M1a–M4).
5. Repeat over 21 window and cutoff settings and over bootstrap resamples to see which structure is stable. Secondary lenses: a locality score (far / near effect) and PC1.
6. As an exploratory check without a lens, cluster with kNN + Louvain and with average / Ward linkage, and measure how well communities track layer (normalised mutual information).

3 · 1-D persistence of distance curves

`exp4/tda3_curves.py`

Goal. Replace eyeballed curve shapes ("decays", "flat", "rises with distance", "spike at the far end") with a count and ranking of each curve's peaks and troughs that survives sampling noise.

Procedure

1. Take each head's $D(d)$ curve for every final token and filler type (1152 curves).
2. Treat the 31 points as a path graph. Compute sublevel-set H0 of f (troughs) and of $-f$ (peaks). A bar's length is the feature's persistence; the global max and min get the curve's range.
3. Build a null: resample the 200 rows 2000 times, subtract the full-sample mean curve, recompute. Set thresholds at the 99th percentile of the null range and of the largest non-essential bar.
4. Call a bar significant if it exceeds its threshold and is at least 0.02 (the repo's margin). No bar counts unless the curve's range passes too.
5. Sort curves into shape classes (near-peak decay, far rise, plateau, ends) and test the five named checks C1–C5 on L9H8, L11H8, L3H0, L0H2 and the previous-token heads. Code was validated against gudhi.

4 · Topology of the attention graph

`exp4/tda4_graph.py`

Goal. Test whether a matching country–capital pair changes the *topology* of the attention network, for example by closing a loop (a persistent 1-cycle) through the capital's edge to its country that a wrong capital does not.

Procedure

1. Re-run the model and save full 33×33 attention patterns for control, own, wrong and random final token, at $\tau \in \{31, 30, 28, 24, 16, 8, 1\}$, both fillers, 200 rows $\times$ 11 countries. Average over the 11 countries, then over rows.
2. Build one graph per layer (mean over its 12 heads) and per named head. Edge weight $w = \max(A_{ij}, A_{ji})$; distance $1 - w$. Delete BOS in the main analysis, since kept it fills every loop.
3. Compute persistent H0 and H1 on the clique complex (gudhi; ripser as a cross-check).
4. Compare own vs wrong: is the difference larger than a swap null (G1), and does only the matching capital close a loop through its country, in every bootstrap (G2)?
5. Check what causes it: transplant the capital→country edge alone and see if the difference disappears (G3a), compare against the BOS-kept graph (B1), and confirm the extraction against Hugging Face.

Verdicts and numbers are in the results one-pager (`topology_onepager.pdf`) and `exp4/LOG.md`. Caveat common to all four: the same 11 pairs throughout, so "beyond the null" means consistent, not large.